SECJ 2203 Software Engineering

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Section 01

NEW PROJECT PROPOSAL

KADA SYSTEM

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1. Introduction

1.1 The Goal

i. The main goal of the proposed system.

The main goal of the proposed KADA Registration System is to create a fully integrated digital platform that enhances the efficiency, security, and user experience for all members and stakeholders of Koperasi Kakitangan KADA Kelantan Berhad. The system seeks to replace the current manual processes, which are time-consuming, error-prone and highly human resource-dependent jobs by replacing present manual processes.

By simplifying and improving their individual responsibilities within the cooperative, members, administrative staff, and Board Directors are the three main user groups that the KADA Registration System benefits. Members may track loan applications, payments, savings, and dividends with real-time updates and automated notifications through an easy-to-use, secure online registration and service portal. The workload and mistakes that come with manual processing are decreased when administrative staff have a centralized platform to monitor and check applications, approve loans, and keep an eye on digital records. The system provides Board Directors with an accountable and transparent means of reviewing important applications and reaching strategic choices that support KADA's objectives. When combined, these enhancements produce a productive, safe, and easy-to-use environment that boosts member happiness and productivity throughout the company.

Ultimately, the goal of the KADA Registration System is to create a safe, effective and user-friendly platform that strengthens security with features, improves data management and facilitates communication. By automating key processes and providing a seamless experience for all users, the system aims to improve service delivery, boost productivity, and foster member satisfaction, driving KADA’s long-term success.

ii. The objectives.

The objective of the KADA Registration System project is to increase productivity, lower human error rates, and improve user experience. The system aims to digitize and streamline the cooperative's member registration, loan application, and financial management procedures. The technologies applied seek to decrease processing times, give members real-time information, and facilitate improved decision-making for board directors and administrative staff by putting automated workflows and safe data handling into place. In the end, this project aims to provide a more effective, accessible, and secure digital platform in order to promote increased productivity and member satisfaction.

1.2 **The Scope**

The scope of our system is to improve the operational efficiency, data accuracy, and user experience for KADA members. By replacing the current manual processes with a digital system, this platform will simplify member registration, loan applications, and financial management in order to increase productivity, lower risk and human error, and cut down on needless extra time spent on paperwork. It is especially beneficial for cooperative members and administrative staff, enabling them to perform their tasks more effectively.

To develop the KADA Registration system, several technologies should be implemented to optimize the system's performance, security and scalability. Each of these technologies has its own value in helping in the development of this system.

No	Technology	Description
1	HTML Programming Language	HTML (HyperText Markup Language) will be used for structuring and coding the content of the KADA Registration System's web interface. Our team is using HTML to design the interface of the website. This programming language provides the foundation for creating and organizing the layout and elements of the user interface, ensuring clear and accessible content display for users.
2	JAVASCRIPT Programming Language	Javascript is used at the backend for this system to handle server-side logic and processing. It is an essential part of any modern website that can make a huge difference on the user experience and functionality. JavaScript enables responsive and efficient server operations, processing member registrations, loan applications, and data requests smoothly.
3	BOOTSTRAP Programming Language (mobile)	Bootstrap is a front-end framework that provides responsive, mobile-first design capabilities. It will be used to create a user-friendly and consistent mobile interface for the KADA system, enabling members to access their accounts and information seamlessly on various devices, including smartphones and tablets.
4	MySQL Database	Relational database management system for structured data storage. It will handle all member records(eg. personal details), their financial information (eg. statements and reports), and loan application data (eg. application status)

		details. MySQL guarantees data integrity, facilitates efficient querying and supports complex data relationships, making it an ideal choice for managing the data needs in the KADA system.

Enables users to view financial statements, track loan balances, and access monthly and annual reports. Supports data analysis for financial trends.

Allows new users to register, input personal details, and securely submit their information. Includes automated status updates and verification.

2	**Loan Management**	Facilitates loan applications, automated eligibility checks, and real-time status tracking for members. Admins can review and approve loan requests efficiently.

2. Software Process Model

2.1 The Goal

The software process model provides a formal structure for organizing and guiding the activities involved in software development, ensuring that the end product meets both the project's objectives and the customer’s requirements. This model describes the steps required for software development, including the tasks to be achieved, inputs and outputs for each work, preconditions and postconditions, and the logical flow between tasks. By building this framework, the software process model aims to simplify the development process and improve collaboration, allowing the team to produce high-quality software more efficiently.

2.2 The Chosen Software Process Model

For the software process model, we selected the waterfall model for the KADA system. This is a linear, sequential life cycle model. It is easy to learn and utilize. In principle, each phase must be completed before proceeding to the next, and no phases overlap.

The first stage is the planning phase. During this phase, our team gathered all of the detailed requirements from our stakeholder, KADA, via online meeting. These thorough criteria involve defining the project scope and gathering information about the existing system's problems through an interview with a KADA representative. After gathering all the information, we documented it, drafted an initial version of the proposal, and submitted it to our lecturer, Dr. Iqbal, for review.

The second stage is the design phase. During this phase, we constructed use case diagrams and the use case descriptions for each module such as applying for membership, and applying for the loan. In addition, we also produced the activity diagram, and sequence diagram and designed the database for each module. Next, we'll use Figma to design a wireframe of the system interface. We then demonstrated our system wireframe to Dr. Iqbal, who checked it and made some suggestions. After completing all of the suggestions, we modified our prototype and submitted the report's final version.

Next is the implantation phase. In this phase, the scratch design KADA system will be implemented into a real prototype design to test the functionality of the new features in the new system. This will involve coding the main features and functionalities of the KADA system. We will use a variety of programming languages such as HTML, CSS, JAVASCRIPT, and

BOOTSTRAP for the front-end interface and MySQL database for the back-end database management. The unit tests will be conducted during the implementation to check the features of the system and ensure that the system has met the customer's requirements.

After the implementation phase is the testing phase. We will produce a user manual and will do a lot of testing such as unit testing, integration testing, system testing and user acceptance testing for the KADA system. The whole system is being tested to evaluate its compliance with its specified requirements. The system will fail if it does not meet the user requirements.

2.3 Project duration

The project will start with requirement analysis, which involves gathering requirements from the client to analyze the existing KADA system. This phase will take 14 days to complete. The deliverable of this stage is the Systems Requirements Specifications (SRS). Next will be the system design phase, which will take around 9 days. It will be divided into two phases. Phase one will involve the design of the system interface and architectural design. After 4 days, the software design for phase two will start and will be evaluated for both the interface and architectural design. For every design stage, we will update another deliverable called the Software Design Document (SDD). Subsequently, after the designing phase is finished, implementation testing will be done for the system, and it will take around 6 weeks to complete. In this phase, the deliverables for this phase include the prototype and unit test results**.** Next, the system testing will begin. This phase will take around 9 days to complete. The deliverables for this stage include the System Testing Document (STD) and the updated prototype.